Quizrr Chapter-Wise Test for JEE Advanced - 2025

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Q.1

An excess of granular zinc was added to $500\ \mathrm{mL}$ of $1\mathrm{M}$ nickel nitrate till equilibrium was established. Find out the concentration of nickel at the equilibrium, if the standard electrode potential of $\mathrm{Zn}^{2+}/\mathrm{Zn}$ and $\mathrm{Ni}^{2+}/\mathrm{Ni}$ are -0.75 and $-0.24\ \mathrm{V}$, respectively. [Given : $\log 1.8 = 0.25$]

(A) $5.56 \times 10^{-18}\mathrm{M}$

(B) $0.56 \times 10^{-18}\mathrm{M}$

(C) $2.16 \times 10^{-18}\mathrm{M}$

(D) $4.12 \times 10^{-18}\mathrm{M}$

Q.2

Consider the cell $\mathrm{Ag(s)|AgBr(s)|Br^-(aq)||AgCl(s)\left|Cl^-(aq)\right|Ag(s)}$ at $25^\circ\mathrm{C}$. The solubility product constants of $\mathrm{AgBr\&AgCl}$ are respectively $5 \times 10^{-13}\&1 \times 10^{-10}$. For what ratio of the concentrations of $\mathrm{Br^-\&Cl^-}$ ions would the emf of the cell be zero ?
[Given: $0.059 \log 5 \times 10^{-13} = -0.7257$]

(A) $1:200$

(B) $1:100$

(C) $1:500$

(D) $200:1$

Q.3

Value of λ_m^0 and λ_m for $0.04\ \mathrm{MCH_3COOH}$ is 390 and 7.8 respectively at 25°C. What is the value $\mathrm{pK_b(CH_3COO^-)}$ at that temperature?

(A) 9.3

(B) 9.2

(C) 4.7

(D) 4.8

Q.4

Equivalent conductivity of $\mathrm{BaCl_2}, \mathrm{H_2SO_4}$ and HCl, are x_1, x_2 and $x_3 \mathrm{Scm^{-1}eq^{-1}}$ at infinite dilution. If conductivity of saturated $\mathrm{BaSO_4}$ solution is $x\mathrm{Scm^{-1}}$, then $\mathrm{K_{sp}}$ of $\mathrm{BaSO_4}$ is :

(A) $\frac{500x}{(x_1+x_2-2x_3)}$

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(B) $\frac{10^6 x^2}{(x_1+x_2-2x_3)^3}$

(C) $\frac{2.5\times10^5 x^2}{(x_1+x_2-x_3)^2}$

(D) $\frac{0.25x^2}{(x_1+x_2-x_3)^2}$

Q.5

A chemist constructed the cell $\mathrm{Mg/Mg(OH)_2(\ s)}$, Buffer $(\mathrm{pH}=?)\ \|\mathrm{Cu^{2+}}\ (10^{-10}\mathrm{M})\ |\ \mathrm{Cu}$. The measured cell voltage was $2.7\ \mathrm{V}$. The left half contains a buffer solution saturated with $\mathrm{Mg(OH)_2}$. The pH in the left half cell is nearly -
(Given SRP values: $\mathrm{Mg^{2+}/Mg}=-2.7\ \mathrm{V}, \mathrm{Cu^{2+}/Cu}=0.3\ \mathrm{V},\ \mathrm{K_{sp}}$ of $\mathrm{Mg(OH)_2}$ is 1.6×10^{-11})

(A) 5

(B) 9

(C) 6

(D) 10

Q.6

Determine potential for the cell $\mathrm{Pt}\left|\begin{matrix}\mathrm{Fe^{+2}}\\ \mathrm{Fe^{+3}}\end{matrix}\right|\left|\mathrm{Cr_2O_7^{-2}, Cr^{+3}, H^+}\right|\mathrm{Pt}$; in which [Fe $^{+2}$] and [Fe $^{+3}$] are $0.5\mathrm{M}$ and $0.75\mathrm{M}$ respectively and $\left[\mathrm{Cr_2O_7^{-2}}\right]$, $\left[\mathrm{Cr^{+3}}\right]$ and $\left[\mathrm{H^+}\right]$are $2\mathrm{M}, 4\mathrm{M}$ and $1\mathrm{M}$ respectively.
Given : $\mathrm{Fe^{+3}+e\rightarrow Fe^{+2}; E^\circ=0.770\ V}$ and $\mathrm{14H^+ + 6e + Cr_2O_7^{-2}\rightarrow 2Cr^{+3}+7H_2O; E^\circ=}$ $1.35\ \mathrm{V}$

(A) $0.12\ \mathrm{V}$

(B) $0.56\ \mathrm{V}$

(C) $0.23\ \mathrm{V}$

(D) $0.92\ \mathrm{V}$

Q.7

Anode oxidation of ammonium hydrogen sulphate produces ammonium persulphate.
$\mathrm{NH_4HSO_4\rightarrow NH_4SO_4^- + H^+}$
$\mathrm{2NH_4SO_4^-\rightarrow [NH_4]_2\ S_2O_8 + 2}e^-$ (Anodic oxidation)
$\mathrm{2H^+ + 2}e^-\rightarrow \mathrm{H_2}$ (cathodic reduction)
Hydrolysis of ammonium persulphate forms $\mathrm{H_2O_2}$:
$\mathrm{[NH_4]_2\ S_2O_8 + H_2O\rightarrow 2NH_4HSO_4 + H_2O_2}$
Current efficiency in the electrolytic process is 60%. Calculate the amount of current required to produce $85\ \mathrm{g}$ of $\mathrm{H_2O_2}$ per hour. Hydrolysis reaction shows 100% yield.

(A) $223.379\ \mathrm{A}$

(B) $323.74\ \mathrm{A}$

(C) $560.43\ \mathrm{A}$

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(D) $300.379\ \text{A}$

Q.8

If SO_2 as an oxidising agent is to be placed on the list of standard reduction potentials, it would be written as $SO_2 + 4H^+ + 4e^- \rightleftharpoons S + 2H_2O$

If the following half-reaction is borne in mind

$$S + 2H^+ + 2e^- \rightleftharpoons H_2\,S\,[E^\circ = 0.14\ \text{V}]$$

Choose a possible E° value for the SO_2 half-reaction.

(A) $-0.70\ \text{V}$

(B) $-0.30\ \text{V}$

(C) $0.16\ \text{V}$

(D) $0.08\ \text{V}$

Q.9

Choose the correct options -

(A) Emf of the cell $\begin{array}{c|c|} \text{Pt}\,(\text{H}_2) & \text{HA} \\ 1\ \text{atm} & \text{pK}_a = 5 \end{array} \Big\| \begin{array}{c|c} \text{HB} & \text{Pt}\,(\text{H}_2) \\ \text{pK}_a = 3 & 1\ \text{atm} \end{array}$

is 0.0591.

(B) The reduction potential for the half cells at 25°C.

$$\text{Mg} \mid \text{Mg}^{2+}\left(1.5 \times 10^{-5}\text{M}\right);\text{E}^\text{o}\text{Mg}, \text{Mg}^{2+} = +2.36\ \text{V is}\ -2.4782\ \text{V}$$

(C) The reduction potential for the half cells at $25^\circ\text{C}.\text{Cl}_2 \mid \text{Cl}^-\left(2 \times 10^{-5}\text{M}\right)$;

$$\text{E}^\text{o}_{\text{Cl}_2,\text{Cl}^-} = +1.36\ \text{V is}\ 1.6377\ \text{V}$$

(D) The reduction potential for the half cells at $25^\circ\text{C}.\text{Pt} \mid \text{Fe}^{2+}(0.1\text{M}), \text{Fe}^{3+}(0.01\text{M})$;

$$\text{E}^\text{o}\text{Fe}^{3+}, \text{Fe}^{2+} = 0.77\ \text{V is}\ 0.2109\ \text{V}$$

Q.10

A current is passed through $500\ \text{mL}$ of an aqueous solution of CaI_2. After sometime, it is observed that 50 mill moles of I_2 have been formed. Which of the following statements is (are) correct?

(A) The number of faradays of charge passed through the solution is $0.10\ \text{F}$.

(B) The volume of dry H_2 at STP that has been formed during electrolysis is $1120\ \text{mL}$

(C) The P^H of the solution is nearly 0.7

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(D) The mass of calcium produced is $2.0\ \mathrm{g}$

Q.11

Choose the correct options -

(A) $KMnO_4$ liberates O_2 from water in presence of an acid.

$$\text{Given, } E^0_{MnO_4^-, Mn^{+2}, H^+|Pt} = 1.51\ \text{V}; E^0_{H^+|O_2|Pt} = 1.223\ \text{V}$$

(B) If the E° for $N_2O_4 + 2H^+ + 2e \rightarrow 2HNO_2$ is $1.07\ \text{V}$. Then the standard cell emf of the cell $2\ N_2O_4 + 2H_2 \rightarrow 4HNO_2$ will be 2.14 volt

(C) At $300\ \text{K}$ specific conductivity of ethanol is $4 \times 10^{-10} \text{mhocm}^{-1}$. The ionic conductance of H^+, $C_2H_5O^-$ at this temperature is 300 and 100mhocm^2 equivalent $^{-1}$ respectively. Then the negative logarithm of ionic product of alcohol will be 18 .

(D) Metallic sodium cannot be prepared from electrolysis of an aqueous solution of $NaCl$ because the SRP $H_2O <$ SRP Na^+.

Q.12

Paragraph:
An electrochemical cell consists of two metallic electrodes dipping in electrolytic solution(s). Thus an important component of the electrochemical cell is the ionic conductor or electrolyte. Electrochemical cells are of two types. In galvanic cell, the chemical energy of a spontaneous redox reaction is converted into electrical work, whereas in an electrolytic cell, electrical energy is used to carry out a non-spontaneous redox reaction. The standard electrode potential for any electrode dipping in an appropriate solution is defined with respect to standard electrode potential of hydrogen electrode taken as zero. The standard potential of the cell can be obtained by taking the difference of the standard potentials of cathode and anode.

$$E^\Theta_{\text{cell}} = E^\Theta_{\text{cathode}} - E^\Theta_{\text{anode}}$$

The standard potential of the cells are related to standard Gibbs energy $\left(\Delta_r G^\Theta = -nFE^\Theta_{\text{cell}}\right)$ and equilibrium constant $\left(\Delta_r G^\Theta = -RT \ln K\right)$ of the reaction taking place in the cell. Concentration dependence of the potentials of the electrodes and the cells are given by Nernst equation.
Question:
The standard emf of a cell, involving one electrons change is found to be $0.591\ \text{V}$ at 25°C. The equilibrium constant of the reaction if $\left(F = 96500 \text{Cmol}^{-1}, R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}\right)$

(A) 1.0×10^1

(B) 1.0×10^{30}

(C) 1.0×10^{10}

(D) 1.0×10^5

Q.13

Paragraph:

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An electrochemical cell consists of two metallic electrodes dipping in electrolytic solution(s). Thus an important component of the electrochemical cell is the ionic conductor or electrolyte. Electrochemical cells are of two types. In galvanic cell, the chemical energy of a spontaneous redox reaction is converted into electrical work, whereas in an electrolytic cell, electrical energy is used to carry out a non-spontaneous redox reaction. The standard electrode potential for any electrode dipping in an appropriate solution is defined with respect to standard electrode potential of hydrogen electrode taken as zero. The standard potential of the cell can be obtained by taking the difference of the standard potentials of cathode and anode.

$$\mathrm{E}_{\text{cell}}^{\Theta} = \mathrm{E}_{\text{cathode}}^{\Theta} - \mathrm{E}_{\text{anode}}^{\Theta}$$

The standard potential of the cells are related to standard Gibbs energy $\left(\Delta_{\mathrm{r}} \mathrm{G}^{\Theta} = -\mathrm{nFE}_{\text{cell}}^{\Theta}\right)$ and equilibrium constant $\left(\Delta_{\mathrm{r}} \mathrm{G}^{\Theta} = -RT \ln \mathrm{K}\right)$ of the reaction taking place in the cell. Concentration dependence of the potentials of the electrodes and the cells are given by Nernst equation.

Question:

The addition of $\mathrm{H_2SO_4}$ to cathode compartment, will -

(A) lower the E and shift equilibrium to the left.

(B) increase the E and shift equilibrium to the left.

(C) increase the E and shift equilibrium to the right.

(D) lower the E and shift equilibrium to the right.

Q.14

Paragraph:

An electrochemical cell consists of two metallic electrodes dipping in electrolytic solution(s). Thus an important component of the electrochemical cell is the ionic conductor or electrolyte. Electrochemical cells are of two types. In galvanic cell, the chemical energy of a spontaneous redox reaction is converted into electrical work, whereas in an electrolytic cell, electrical energy is used to carry out a non-spontaneous redox reaction. The standard electrode potential for any electrode dipping in an appropriate solution is defined with respect to standard electrode potential of hydrogen electrode taken as zero. The standard potential of the cell can be obtained by taking the difference of the standard potentials of cathode and anode.

$\mathrm{E}_{\text{cell}}^{\Theta} = \mathrm{E}_{\text{cathode}}^{\Theta} - \mathrm{E}_{\text{anode}}^{\Theta}$

The standard potential of the cells are related to standard Gibbs energy $\left(\Delta_{\mathrm{r}} \mathrm{G}^{\Theta} = -\mathrm{nFE}_{\text{cell}}^{\Theta}\right)$ and equilibrium constant $\left(\Delta_{\mathrm{r}} \mathrm{G}^{\Theta} = -RT \ln \mathrm{K}\right)$ of the reaction taking place in the cell. Concentration dependence of the potentials of the electrodes and the cells are given by Nernst equation.

Question:

For a cell reaction involving two electron change, the standard EMF of the cell is $0.295\ \mathrm{V}$ at $25^{\circ}\mathrm{C}$. The equilibrium constant of the reaction at $25^{\circ}\mathrm{C}$ will be:

(A) 1×10^{-10}

(B) 2.95×10^{-2}

(C) 10

(D) 1×10^{10}

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Q.15

Paragraph:

A Zn/Cu galvenic cell was taken at standard state condition. Now the cell is allowed to discharge spontaneously till the concentration in cathodic and anodic compartment changes by 20% of the initial value. Given : $E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V, E^{\circ}_{Cu^{2+}/Cu} = +0.34\ V$

At $300\ K$, $\frac{2.303RT}{F} = 0.06 volt$

Use : $\ln 3 = 0.48, \ln 2 = 0.8$

All data at $300\ K$.

Question:

The conc. of Zn^{2+} after discharge is

(A) $1.2M$

(B) $1.8M$

(C) $0.8M$

(D) $0.6M$

Q.16

Paragraph:

A Zn/Cu galvenic cell was taken at standard state condition. Now the cell is allowed to discharge spontaneously till the concentration in cathodic and anodic compartment changes by 20% of the initial value. Given : $E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V, E^{\circ}_{Cu^{2+}/Cu} = +0.34\ V$

At $300\ K$, $\frac{2.303RT}{F} = 0.06 volt$

Use : $\ln 3 = 0.48, \ln 2 = 0.8$

All data at $300\ K$.

Question:

What is the change in emf of the cell after the discharge (in volt) -

(A) 2.7×10^{-3}

(B) $+5.4 \times 10^{-3}$

(C) -5.4×10^{-3}

(D) 1.08×10^{-2}

Q.17

What is the change in ΔG during the discharge process at $(300\ K)(J) -$

(A) -248.72

(B) - 1033.9

(C) $+248.72$

(D) $+449$

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Q.18

Match the column

Column I	**Column II**
(A) $Zn \mid Zn^{2+}$ (c) $\parallel Zn^{2+}$ (2C) $\mid Zn$	(p) Spontaneous cell reaction
(B) H_2 (P = 1 atm) $\mid HCl$ (1N) $\parallel H_2SO_4$ (1N) $\mid H_2$ (P =1 atm)	(q) Working cell representation
(C) $Cu \mid Cu^{2+}$ (0.01 M) $\parallel Ag^+$ (0.1 M) $\mid Ag$	(r) Concentration of cation in cathodic compartment increases upto equilibrium
(D) $Ag \mid AgCl \mid (KCl, 0.1\ M) \parallel Ag^+$ (0.01 M) $\mid Ag$	(s) Concentration cell

(A) A-p,q,s; B-r,s ; C- p,q ; D-p,q,s;

(B) A-p,q; B-r ; C- p,q,s ; D-p,q,s;

(C) A-p,q,s; B-p,r ; C- p,q,r,s ; D-p,q,s;

(D) A-r,s; B-p,r ; C- p ; D-p,q,s;

Q.19

A fuel cell uses $CH_4(g)$ and forms CO_3^{2-} at the anode. It is used to power a car with 80 amp. for $0.96 hr$. How many litres of $CH_4(g)(STP)$ would be required ?
$(V_m = 22.4\ L/mol)\ (F = 96500)$. Assume 100% efficiency. (round off to nearest integer)

Q.20

The standard reduction potential $E_{Bi^{3+}/Bi}$ and $E^{\circ}_{Cu^{2+}}/Cu$ are 0.212 and $0.33\ V$ respectively. A mixture of salt of Bi^{3+} and Cu^{2+} at unit concentration each is electrolyzed at $25^{\circ}C$. To what value can $[Cu^{2+}]$ be brought down before bismuth start to deposit it during electrolysis ? Give your answer in the multiple of 10^{-4}.

Answers & Solutions

Q.1 Answer:

—

Solution:

The equation for cell reaction is :

$$Zn(s) + Ni(NO_3)_2 \rightarrow Zn(NO_3)_2 + Ni(s)$$
$$\text{or} \quad Zn(s) + Ni^{2+}(aq) \rightarrow Zn^{2+}(aq) + Ni(s)$$

In this cell zinc acts as anode while nickel acts as cathode so $E_{cell} = E_{cathode} - E_{anode}$ (on the basis of SRP)

$$E_{cell} = -0.24 - (-0.75) = +0.51\ V$$

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Nernst equation for above reaction

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{n} \log \frac{[Zn^{2+}]}{[Ni^{2+}]}$$

At equilibrium $E_{cell} = 0$ **and** $n = 2$

So, $0 = 0.51 - \frac{0.0591}{2} \log \frac{[Zn^{2+}]}{[Ni^{2+}]}$

or $\log \frac{[Zn^{2+}]}{[Ni^{2+}]} = \frac{0.51 \times 2}{0.0591} = 17.25$ **or** $\frac{[Zn^{2+}]}{[Ni^{2+}]} = 1.8 \times 10^{17}$

Let $x mol L^{-1}$ **be the concentration of** $Ni^{2+}(aq)$ **at equilibrium.**

So $Zn^{2+}(aq)$ **ion conc.** $= (1 - x) mol L^{-1} \approx 1 \text{ mol } L^{-1}$

So $\frac{1}{x} = 1.8 \times 10^{17}$ **or** $x = \frac{1}{1.8 \times 10^{17}} = 5.56 \times 10^{-18} M$

Hence, $[Ni^{2+}] = 5.56 \times 10^{-18} M$

Q.2 Answer:

$1 : 200$

Solution:

$$E^{0}_{Br^-/AgBr/Ag} = E^{0}_{Ag^+/Ag} + \frac{0.059}{1} \log K_{SP}$$

$$AgBr = E^{0}_{Ag^+/Ag} - 0.7257$$
$$\text{and } E^{0}_{Cl^-/AgCl/Ag} = E^{0}_{Ag^+/Ag} + \frac{0.059}{1} \log K_{SP}$$
$$AgCl = E^{0}_{Ag^+/Ag} - 0.59$$

Now cell reaction is

$$Ag + Br^- \rightarrow AgBr + 1e^-$$
$$AgCl + 1e^- \rightarrow Ag + Cl^-$$

$$Br^- + AgCl \xrightarrow{le^-} Cl^- + AgBr$$
$$0 = (0.7257 - 0.59) + \frac{0.059}{1} \log \frac{[Br^-]}{[Cl^-]} \Rightarrow \frac{[Br^-]}{[Cl^-]} = 0.005$$

Q.3 Answer:

9.2

Solution:

$\alpha \rightarrow$ **Degree of dissociation**

$\alpha = \frac{\lambda_m^{\circ}}{\lambda m}$

$\alpha = \frac{7.8}{390} = \frac{1}{50} (\approx 2\%)$

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$(Ka)_{CH_3COOH} = \frac{c\alpha^2}{1-\alpha}\ 1-\alpha = 1\ (\alpha \approx 2\%)$

$= C\alpha^2$

$= 0.04\ \times\ \frac{1}{2500}$

$K_a = 1.6 \times 10^{-5}$

$P^{k_a} = 6 - log16$

$= 6 - 4 \times 0.3 = 4.8$

$P^{k_a} + P^{k_b}$ **=14**

$P^{k_b} = 14 - 4.8 = 9.2$

Q.4 Answer:

$\frac{2.5\times10^5 x^2}{(x_1+x_2-x_3)^2}$

Solution:

$$\Lambda_m^\infty\left(\mathrm{BaSO_4}\right) = 2\Lambda_{eq}^\infty\left(\mathrm{BaSO_4}\right)$$
$$\Lambda_m^\infty\left(\mathrm{BaSO_4}\right) = \Lambda_{\mathrm{eq}}^\infty\left(\mathrm{Ba^{2+}}\right) + \Lambda_{eq}^\infty\left(\mathrm{SO_4^{2-}}\right)$$
$$= \Lambda_{\mathrm{eq}}^\infty\left(\mathrm{BaCl_2}\right) + \Lambda_{\mathrm{eq}}^\infty\left(\mathrm{H_2SO_4}\right) - \Lambda_{\mathrm{eq}}^\infty\left(\mathrm{HCl}\right)$$
$$\Lambda_{\mathrm{eq}}^\infty\left(\mathrm{BaSO_4}\right) = x_1 + x_2 - x_3$$
$$\Lambda_{\mathrm{eq}}^\infty = 2\left(x_1 + x_2 - x_3\right)$$

For sparingly soluble salt

$$\Lambda_m^\infty = \frac{k}{M} \times 1000$$
$$M = \frac{x}{2\left(x_1 + x_2 - x_3\right)} \times 1000$$
$$= \frac{500x}{\left(x_1 + x_2 - x_3\right)}$$
$$\therefore \quad K_{sp} = M^2$$
$$= \frac{2.5 \times 10^5 x^2}{\left(x_1 + x_2 - x_3\right)^2}$$

So, (c) is correct.

Q.5 Answer:

9

Solution:

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$$Mg \longrightarrow Mg^{2+} + 2e^-$$

$$Cu^{2+} + 2e^- \longrightarrow Cu$$
$$Mg(s) + Cu^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Cu(s)$$
$$E_{cell} = E^\circ_{cell} - \frac{0.059}{2} \log_{10} \frac{[Mg^{2+}]}{[Cu^{2+}]}$$
$$\text{So, } \frac{[Mg^{2+}]}{[Cu^{2+}]} \approx 10^{10} \text{ or } [Mg^{2+}] = 1$$
$$\text{So, } K_{sp} = [Mg^{2+}][OH^-]^2 \text{ or } [OH^-]^2 = 16 \times 10^{-12}$$
$$\text{So, } pOH = 6 - \log 4 = 5.4 \text{ or } pH = 14 - 5.4 = 8.6 \approx 9$$

Q.6 Answer:

0.56 V

Solution:

$E^oFe^{+2}/Fe^{+3} > E^o_{Cr}/Cr_2^{+6}$

Redox changes will be

Anode : $Fe^{+2} \rightarrow Fe^{+3} + e$

Cathode : $14H^+ + 6e + Cr_2O_7^{-2} \rightarrow 2Cr^{+3} + 7H_2O$

$$E^\circ_{cell} = E^oFe^{+2}/Fe^{+3} - E^o_{Cr} + 3/Cr_2^{+6} = -0.770 + 1.35 = 0.58 \text{ V}$$
$$E_{cell} = E^o_{cell} - \frac{0.059}{6} \log \frac{[Fe^{+3}]^6[Cr^{+3}]^2}{[Fe^{+2}]^6[Cr_2O_7^{-2}][H^+]^{14}}$$

$$= 0.58 - \frac{0.059}{6} \log \frac{(0.75)^6(4)^2}{(2)(1)^{14}(0.5)^6}$$
$$= 0.58 - \frac{0.059}{6} \log(91.125) = 0.58 - 0.0192 = 0.56 \text{ V}$$

Q.7 Answer:

223.379 A

Solution:

$$(NH_4)_2\ S_2O_8 + H_2O \rightarrow 2NH_4HSO_4 + H_2O_2$$
$$228 \text{ g} \qquad 34 \text{ g}$$
$34 g H_2O_2$ is produced by 228 g $[NH_4]_2\ S_2O_8$

$85\ g H_2O_2$ **will be produced by** $\frac{228}{34} \times 85$ g $(NH_4)_2\ S_2O_8 = 570$ g

Equivalent mass of $(NH_4)_2\ S_2O_8$ **may be calculated using the following reactions:**

$2NH_4SO_4^- \rightarrow [NH_4]_2\ S_2O_8 + 2e^-$

Equivalent mass of $[NH_4]_2\ S_2O_s = \frac{\text{m.mass}}{2} = \frac{228}{2} = 114$

The no. of g equivalents of $(NH_4)_2\ S_2O_8$ **required** $= \frac{570}{114}$

From Faraday's laws of electrolysis, $I = \frac{Q}{t} = \frac{570 \times 96500}{114 \times 3600} = 134$ A

Given that current efficiency is 60%**, the actual amount of current** $= \frac{114}{60} \times 134 = 223.379$ A

Q.8 Answer:

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0.16 V

Solution:

For the reaction: $SO_2 + H_2S \rightarrow 2S + 2H_2O$

$$E^\circ_{cell} = E^\circ_{OA} - E^\circ_{RA} = E^\circ_{S/SO_2} - 0.14 > 0 \quad \therefore \quad E_{S/SO_2} > 0.14$$

Q.9 Answer:

Emf of the cell $\begin{vmatrix} Pt\,(H_2) \\ 1\text{ atm} \end{vmatrix} \begin{vmatrix} HA \\ pK_a = 5 \end{vmatrix} \|$

$$\begin{vmatrix} HB \\ pK_a = 3 \end{vmatrix} \begin{vmatrix} Pt\,(H_2) \\ 1\text{ atm} \end{vmatrix}$$

is 0.0591.

Solution:

(A) Left half cell $E_{H/H^+} = E^\circ_{oxid} - \frac{0.0591}{1}\log[H]$

$E_{H/H^+} = E^o_{oxid} - 0.0591pH(HA)$

Right half cell $E_{(H^+/H)} = E^o_{Red} - 0.0591\log\frac{1}{[H^+]_{HB}}$

$E_{(H^+/H)} = E^o_{Red} - 0.0591pH(HB)$

For HA $(H^+) = \sqrt{CK_a}$ or $-\log[H^+] = -\frac{1}{2}\log C - \frac{1}{2}\log K_a$

$(pH)_{HA} = \frac{1}{2}(pK_a)_{HA} - \frac{1}{2}\log C$

Similarly for HB : $(pH)_{HB} = \frac{1}{2}(pK_a)_{HB} - \frac{1}{2}\log C$

$E_{cell} = E_{right} - E_{left} = 0.0591\left[\frac{1}{2}pKa_{(HA)} - \frac{1}{2}pKa_{(HB)}\right] = \frac{0.0591}{2}[5-3] = 0.0591.$

(B)

$Mg^{2+} + 2e \rightleftharpoons Mg$ **(reduction)**

$$E^{2+}_{Mg}, Mg = E^o_{Mg^{2+},Mg} + \frac{0.0591}{2}\log\left(1.5\times10^{-5}\right) = -2.36 - \frac{4\times0.0591}{2} = -2.4782\text{ V}$$

(C)

$1/2Cl_2 + e \rightleftharpoons Cl-$ **(reduction)**

$$E_{Cl_2,Cl^-} = E^o_{Cl_2,Cl^-} - \frac{0.0591}{1}\log\left[Cl^-\right] = 1.36 - \frac{0.0591}{1}\log\left(2\times10^{-5}\right) = 1.6377\text{ V}$$

(D)

$Fe^{3+} + e \rightleftharpoons Fe^{2+}$ **(reduction)**

$$E^oFe^{3+}, Fe^{2+} = E^oFe^{3+}, Fe^{2+} - \frac{0.0591}{n}\log\frac{\left[Fe^{2+}\right]}{\left[Fe^{3+}\right]}$$

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$$= 0.77 - \frac{0.0591}{1} \log \frac{0.1}{0.01} = 0.7109 \text{ V}.$$

Hence, options (A), (B), (C) are correct.

Q.10 Answer:

The number of faradays of charge passed through the solution is 0.10 F.

Solution:

$\mathrm{CaI_2}(aq)$: **Cathode:** $\mathrm{Ca}^{2+} \quad H^{\oplus}$

Anode: $I^{\circ} \quad \overset{\circ}{O}\ H$

$$\left[\begin{array}{c} \text{Mmoles } I_2 = 50 \Rightarrow 100\text{mF passed } = 0.1 \text{ F} \\ (2I^{\circ} \rightarrow I_2 + 2e^{-}) \end{array}\right]$$

b.Also, mmoles of $H_2 = \frac{1}{2} \times$ number of mF

$$= \frac{1}{2} \times 100 = 50 \Rightarrow V_{H_2 \text{ at } STP}$$

$$= 50 \times 22.4\text{ml} = 1120 \text{ mL}$$

c. 100 mmoles OH **left**

$$\Rightarrow [\overset{\circ}{O}H] = \frac{100}{500} = 0.2$$

$$\Rightarrow P^{OH} = -\log 0.2 = 0.7$$

$$\Rightarrow P^{H} = 13.3$$

Q.11 Answer:

$\mathrm{KMnO_4}$ liberates $\mathrm{O_2}$ from water in presence of an acid.

$$\text{Given, } \mathrm{E^0_{MnO_4^-, Mn^{+2}, H^+|Pt}} = 1.51 \text{ V}; \mathrm{E^0_{H^+|O_2|Pt}} = 1.223 \text{ V}$$

Solution:

(A) The cell for oxidation reaction is

$$\mathrm{Pt\,|O_2|\,H^+, H_2O \| MnO_4^-, Mn^{+2}, H^+\ |\ Pt}$$

$$\therefore \quad \mathrm{E^0_{cell}} = \mathrm{E^0_{MnO_4^0, Mn^{+2}, H^+|Pt}} - \mathrm{E^0_{H^+|O_2|Pt}} = 1.51 \text{ V} - 1.223 \text{ V} = 0.287 \text{ V}$$

$$(\mathrm{B}) \quad \mathrm{2\,N_2O_4 + 2H^+ + 2e \rightarrow 2HNO_2}$$

$$\mathrm{H_2 \rightarrow 2H^+ + 2e}$$

$$\mathrm{N_2O_4 + H_2 \rightarrow 2HNO_2}$$

$$\mathrm{E^0_{cell}} = \mathrm{E^0_{N_2O_4|H^+, HNO_2}} + \mathrm{E^0_{H_2|2H^+}} = 1.07 \text{ V}$$

$$\mathrm{C)} \quad \lambda^{\infty} = \lambda^{\infty}\mathrm{H^{+}}^{+} + \lambda^{\infty}\mathrm{C_2H_5O^-} = 400$$

$$\lambda = \mathrm{K} \times \frac{1000}{\mathrm{C}} \left[\mathrm{C} = \text{ concentration of } \mathrm{C_2H_5OH}\right]$$

$$\mathrm{C} = \frac{4 \times 10^{-10} \times 1000}{400}$$

$$\mathrm{K_{alcohol}} = \left[\mathrm{H^+}\right]\left[\mathrm{OC_2H_5}\right] = \left(10^{-9}\right)^2$$

$$\mathrm{pK_{alcohol}} = -\log\left(10^{-18}\right) = 18$$

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(C) $\lambda^{\infty}=\lambda_{H^{+}}^{\infty}+\lambda_{C_2H_5O^{-}}^{\infty}=400$

(D) SRP H_2O **>SRP** Na^{+} **so** H_2O **will be reduced in preference to** Na^{+} **ion.**

Q.12 Answer:

1.0×10^{10}

Solution:

Equilibrium constant is related to the cell emf as follows:

$$K=10^{\left(nE_{cell}^{0}/0.0591\right)}$$

For the given cell, $K=10^{(0.591)/0.0591}=10^{10}=1.0 \times 10^{10}$

Q.13 Answer:

increase the E and shift equilibrium to the right.

Solution:

For the cathode,

The Nernst equation for the H^{+}/H_2 **half cell (at** $298\ K$ **) is**

$$E=0-\frac{0.0591}{2}\log\frac{1}{\left[H^{+}\right]^2}=0.0591\log\left[H^{+}\right]$$

The emf of the cathode half-cell will increase on the addition of H_2SO_4

This will facillate the forward reaction.

Q.14 Answer:

1×10^{10}

Solution:

At $298\ K, E_{cell}^{0}=\frac{0.0591}{2}\log K$

$\log K=\frac{2\times E_{cell}^{0}}{0.0591}=\frac{2\times 0.295}{0.0591}=10; K=1\times 10^{10}$

Q.15 Answer:

$1.2M$

Solution:

(i) The cell $Zn(s)\left|Zn^{2+}(1M)\right|\left|Cu^{2+}(1M)\right|Cu(s)$

$\frac{Cu^{2+}+2e-Cu(s)}{Zn(s)+Cu^{2+}(aq)\longrightarrow Cu(s)}$

The conc. of Zn^{2+} **after discharge**

$\left[Zn^{2+}\right]=1+0.2=1.2M\ \left[Cu^{2+}\right]=1-0.2=0.8M$ **Now,** $E=E^{\circ}-\frac{0.06}{2}\log\left[\frac{1.2}{0.8}\right]\ E=$

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$E^\circ - (0.03)\log\left(\frac{3}{2}\right)\ E = E^\circ - 0.03[0.48 - 0.30]\ E = E^\circ - (0.03)(0.18)$

$$\left[Zn^{2+}\right] = 1 + 0.2 = 1.2M$$
$$\left[Cu^{2+}\right] = 1 - 0.2 = 0.8M$$
$$\text{Now, } E = E^\circ - \frac{0.06}{2}\log\left[\frac{1.2}{0.8}\right]$$
$$E = E^\circ - (0.03)\log\left(\frac{3}{2}\right)$$
$$E = E^\circ - 0.03[0.48 - 0.30]$$
$$E = E^\circ - (0.03)(0.18)$$

Chang in EMF $= -0.0054$ **volt**

(ii) Change in ΔG **in the process**

$$E = E^\circ - \frac{RT}{nF}\ln Q$$
$$-nFE = -nFE^\circ + RT\ln Q$$
$$\Delta G = \Delta G^\circ + RT\ln Q$$
$$\Delta G = \Delta G^\circ + 8.314 \times 300 \times 0.18$$
$$\Delta G - \Delta G^\circ = +8.314 \times 300 \times 0.18 = +449 \text{ Joule.}$$

Q.16 Answer:

-5.4×10^{-3}

Solution:

(i) The cell $Zn(s)\left|Zn^{2+}(1M)\right|\left|Cu^{2+}(1M)\right|Cu(s)$

$\frac{Cu^{2+} + 2e - Cu(s)}{Zn(s) + Cu^{2+}(aq) \longrightarrow Cu(s)}$

The conc. of Zn^{2+} **after discharge**

$\left[Zn^{2+}\right] = 1 + 0.2 = 1.2M\ \left[Cu^{2+}\right] = 1 - 0.2 = 0.8M$ **Now,** $E = E^\circ - \frac{0.06}{2}\log\left[\frac{1.2}{0.8}\right]\ E =$
$E^\circ - (0.03)\log\left(\frac{3}{2}\right)\ E = E^\circ - 0.03[0.48 - 0.30]\ E = E^\circ - (0.03)(0.18)$

$$\left[Zn^{2+}\right] = 1 + 0.2 = 1.2M$$
$$\left[Cu^{2+}\right] = 1 - 0.2 = 0.8M$$
$$\text{Now, } E = E^\circ - \frac{0.06}{2}\log\left[\frac{1.2}{0.8}\right]$$
$$E = E^\circ - (0.03)\log\left(\frac{3}{2}\right)$$
$$E = E^\circ - 0.03[0.48 - 0.30]$$
$$E = E^\circ - (0.03)(0.18)$$

Chang in EMF $= -0.0054$ **volt**

(ii) Change in ΔG **in the process**

$$E = E^\circ - \frac{RT}{nF}\ln Q$$
$$-nFE = -nFE^\circ + RT\ln Q$$
$$\Delta G = \Delta G^\circ + RT\ln Q$$
$$\Delta G = \Delta G^\circ + 8.314 \times 300 \times 0.18$$
$$\Delta G - \Delta G^\circ = +8.314 \times 300 \times 0.18 = +449 \text{ Joule.}$$

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Q.17 Answer:

$+449$

Solution:

(i) The cell $\mathrm{Zn(s)\left|Zn^{2+}(1M)\right|\left|Cu^{2+}(1M)\right|Cu(s)}$

$\frac{\mathrm{Cu^{2+}+2e-Cu(s)}}{\mathrm{Zn(s)+Cu^{2+}(aq)\longrightarrow Cu(s)}}$

The conc. of $\mathrm{Zn^{2+}}$ **after discharge**

$\left[\mathrm{Zn^{2+}}\right]=1+0.2=1.2\mathrm{M}\ \left[\mathrm{Cu^{2+}}\right]=1-0.2=0.8\mathrm{M}$ **Now,** $\mathrm{E=E^{\circ}-\frac{0.06}{2}\log\left[\frac{1.2}{0.8}\right]\ E=E^{\circ}-(0.03)\log\left(\frac{3}{2}\right)\ E=E^{\circ}-0.03[0.48-0.30]\ E=E^{\circ}-(0.03)(0.18)}$

$$\begin{aligned}&\left[\mathrm{Zn^{2+}}\right]=1+0.2=1.2\mathrm{M}\\&\left[\mathrm{Cu^{2+}}\right]=1-0.2=0.8\mathrm{M}\\&\text{Now, }\mathrm{E=E^{\circ}-\frac{0.06}{2}\log\left[\frac{1.2}{0.8}\right]}\\&\mathrm{E=E^{\circ}-(0.03)\log\left(\frac{3}{2}\right)}\\&\mathrm{E=E^{\circ}-0.03[0.48-0.30]}\\&\mathrm{E=E^{\circ}-(0.03)(0.18)}\end{aligned}$$

Chang in EMF $=-0.0054$ **volt**

(ii) Change in $\Delta\mathrm{G}$ **in the process**

$$\begin{aligned}&\mathrm{E=E^{\circ}-\frac{RT}{nF}\ln Q}\\&\mathrm{-nFE=-nFE^{\circ}+RT\ln Q}\\&\mathrm{\Delta G=\Delta G^{\circ}+RT\ln Q}\\&\mathrm{\Delta G=\Delta G^{\circ}+8.314\times300\times0.18}\\&\mathrm{\Delta G-\Delta G^{\circ}=+8.314\times300\times0.18=+449\ Joule.}\end{aligned}$$

Q.18 Answer:

A-p,q,s; B-r,s ; C- p,q ; D-p,q,s;

Solution:

((A) - p,q,s, (B) - r,s, (C) - p,q, (D) - p,q,s)

(A) $\mathrm{E_{cell}=E^{\circ}_{cell}-\frac{0.0591}{2}\log\frac{\left[Zn^{2+}\right]\text{ anode}}{\left[Zn^{2+}\right]\text{ cathode}}}$

At anode $\mathrm{Zn\rightarrow Zn^{2+}+2e^{-}}$

At cathode $\mathrm{Zn^{2+}+2e^{-}\rightarrow Zn}$

$\mathrm{E_{cell}=0-\frac{0.0591}{2}\log\frac{C_1}{C_2}}$

$\Delta\mathrm{G_1}=-nFE_{\mathrm{cell}}$

for $\Delta\mathrm{G_1}=-\mathrm{ve}$

$E_{\mathrm{cell}}=+\mathrm{ve}$

$\Rightarrow C_2>C_1$

since $2c>c$ **this holds**

(B) $\mathrm{H_2SO_4}$ **compartment acts as anode**

(C) $\mathrm{E=+ve,\Delta G=-ve}$

(D) $\mathrm{Ag^{+}(0.01M)\rightarrow Ag^{+}\left(K_{sp}/0.1\right),E=+ve}$

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Q.19 Answer:

—

Solution:

$CH_4 + 10OH^- \longrightarrow CO_3^{2-} + 7H_2O + 8e^-$

No. of Faradays required $= \frac{80 \times 3600 \times 9.8}{96500}$

Hence, mol of CH_4 **required** $= \frac{1}{8} \times \frac{80 \times 3600 \times 9.8}{96500}$

$$V_{CH_4} = \frac{1}{8} \times \frac{80 \times 3600 \times 9.8}{96500} \times 22.4\ L = 8.356 \times 0.96 = 8.02\ L$$

Q.20 Answer:

—

Solution:

$E^\circ_{Cu^2//Cu} = 0.33\ V \& E^\circ_{Bi^{3+}/Bi} = 0.212\ V$

On passing current initially Cu^{2+} **deposits till** $E_{Cu^{2+}}/Cu$ **become** $0.212\ V$**, then** Bi^{3+} **will be deposited.**

$$E^\circ_{Bi^{3+}/Bi} = E^\circ_{Cu^{2+}/Cu} - \frac{0.059}{2} \log \frac{1}{[Cu^{+2}]}$$
$$0.0212 = 0.33 - \frac{0.059}{2} \log \frac{1}{[Cu^{+2}]}$$
$$\log \frac{1}{[Cu^{+2}]} = 4$$
$$\therefore \quad [Cu^{2+}] = 10^{-4} M$$
$$[Cu^{2+}] \times 10^{-4} = 1$$

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